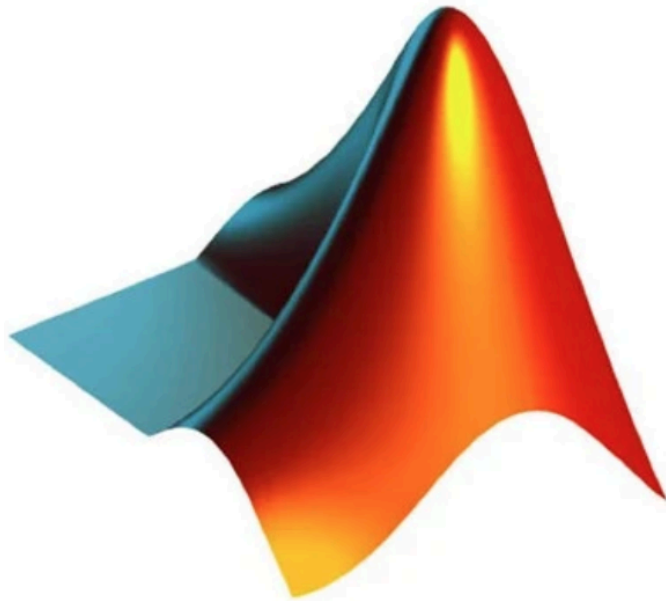
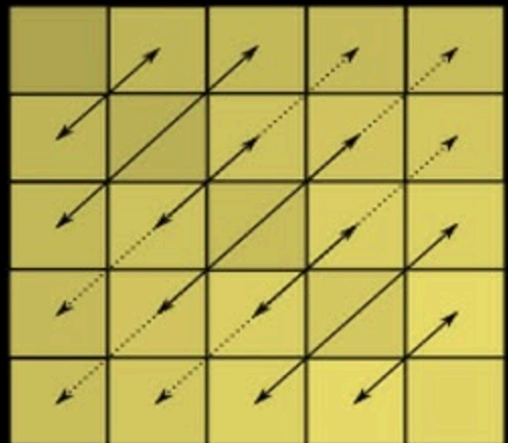


MATH 284 MATLAB Project #6 - Symmetric Matrix & Quadratic Form Studio



MATLAB®

$$A^T = A$$



Project Objective

Concepts And Textbook Chapters/Sections Covered

Chapter 7: Symmetric Matrices and Quadratic Forms,

- 7.1 Diagonalization of Symmetric Matrices
- 7.2 Quadratic Form

- Symmetric matrices
- Orthogonal diagonalization
- Spectral decomposition
- Eigenvalues and eigenvectors of symmetric matrices
- Quadratic forms
- Principal axes
- Positive/Negative Definiteness
- Matrix classification

GUI Layout

Symmetric Matrix & Quadratic Form Studio	
Matrix Input	
Matrix A	
[	]
[Analyze Matrix]	
Matrix Properties	
Symmetric?	
Eigenvalues	
Eigenvectors	
Positive Definite?	
Rank	
Orthogonal Diagonalization	
Matrix Q	
Matrix D	
Verification	
Quadratic Form	
Formula	
Classification	
Principal Axes	
Visualization	
Contour Plot	
Surface Plot	
Principal Axes	
Applications	
[Energy Surface]	
[Optimization]	
[Conic Section Viewer]	

Necessary MATLAB Files For This Project

SymmetricMatrixStudioApp.m

Responsibilities:

- *Build GUI*
- *Coordinate computations*
- *Display visualizations*
- *Manage user interaction*

SymmetricMatrixAnalyzer.m

Stores:

- *the matrix A*
- *if the matrix is symmetric*
- *eigenvalues and eigenvectors*
- *rank*
- *determinant*

Displays:

- *if matrix is symmetric*
- *eigenvalues*
- *orthogonal eigenvectors*
- *positive definite*

OrthogonalDiagonalization.m

Computes $Q^T A Q = D$

Features:

- *orthogonal matrix Q*
- *diagonal matrix D*
- *verification of the above theorem*

QuadraticFormAnalyzer.m

Computes $Q(x) = x^T A x$

Analyzes:

- *Positive definite*
- *Negative definite*
- *Positive semidefinite*
- *Negative semidefinite*
- *Indefinite*

QuadraticFormVisualizer.m

Displays:

- *Contour maps*
- *3D surfaces*
- *Principal axes*
- *Eigenvector directions*

DefinitenessAnalyzer.m

Determines if the matrix is:

- Positive definite*
- Positive semidefinite*
- Negative definite*
- Negative semidefinite*
- Indefinite*

Necessary MATLAB Methods For This Project

createComponents()

analyzeMatrix()

diagonalize()

classifyQuadraticForm()

updatePlots()

clearGUI()

checkSymmetry()

computeEigenvalues()

computeEigenvectors()

classifyMatrix()

computeQ()

computeD()

`verifyDiagonalization()`
`evaluateQuadraticForm()`
`classify()`
`principalAxes()`
`plotContour()`
`plotSurface()`
`plotPrincipalAxes()`
`animateRotation()`
`isPositiveDefinite()`
`isNegativeDefinite()`
`classifyEigenvalues()`

GUI Functions

- **Orthogonal diagonalization explorer**
- **Quadratic form analyzer and visualizer**

Pseudocode

START PROGRAM

Create GUI

Initialize

Matrix input

Results panel

Visualization panel

WHEN Analyze Matrix pressed

Read matrix

Check symmetry

Compute eigenvalues

Compute eigenvectors

Determine definiteness

Display results

WHEN Orthogonal Diagonalization selected

Compute

Q

D

Verify

$$\mathbf{Q}^T \mathbf{A} \mathbf{Q} = \mathbf{D}$$

WHEN Quadratic Form selected

Compute

$$\mathbf{x}^T \mathbf{A} \mathbf{x}$$

Classify

Plot contour

Plot surface

WHEN Applications selected

Load engineering examples

WHEN Clear selected

Reset interface

END PROGRAM

$$\begin{bmatrix} 0 & 1 & 8 & 9 \\ 1 & 9 & 7 & 3 \\ 9 & 8 & 2 & 5 \\ 1 & 7 & 3 & 8 \end{bmatrix} \quad \begin{array}{ccc} \text{Row A} & \text{Col A} & \text{Nul A} \end{array} \quad \begin{bmatrix} 5 & 8 & 1 & 0 \\ 2 & 3 & 5 & 0 \\ 9 & 8 & 0 & 0 \\ 6 & 7 & 2 & 4 \end{bmatrix}$$

linear algebra

$\lambda_1 = 3 \quad \lambda_2 = 7 \quad \det(A) = 0$